

Theorem (Chebyshev's Sum Inequality). *Let*

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n$$

be real numbers. Then

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right). \quad (1)$$

Proof. Degenerate cases. If $n = 1$ the inequality reduces to $a_1 b_1 \geq a_1 b_1$, true with equality. If all the a_i are equal (or all the b_i are equal) both sides of (1) equal $a_1 \sum_i b_i$ (respectively $b_1 \sum_i a_i$), so equality holds. These situations will be mentioned again at the end of the proof.

Reformulation of the inequality. The right-hand side of (1) can be written as a double sum:

$$\left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) = \sum_{i=1}^n a_i b_i + \sum_{1 \leq i < j \leq n} (a_i b_j + a_j b_i). \quad (2)$$

Subtracting $\sum_i a_i b_i$ from both sides of (1) and using (2) we obtain the equivalent statement

$$(n-1) \sum_{i=1}^n a_i b_i \geq \sum_{1 \leq i < j \leq n} (a_i b_j + a_j b_i). \quad (3)$$

A non-negative elementary inequality. For any pair $i < j$ the monotonicity $a_i \leq a_j$ and $b_i \leq b_j$ implies

$$(a_j - a_i)(b_j - b_i) \geq 0. \quad (4)$$

Expanding the product gives

$$a_j b_j + a_i b_i - a_j b_i - a_i b_j \geq 0,$$

or equivalently

$$a_i b_i + a_j b_j \geq a_i b_j + a_j b_i. \quad (5)$$

Summation over all unordered pairs. Summing (5) over all $\binom{n}{2}$ pairs (i, j) with $i < j$ yields

$$\sum_{1 \leq i < j \leq n} (a_i b_i + a_j b_j) \geq \sum_{1 \leq i < j \leq n} (a_i b_j + a_j b_i). \quad (6)$$

Each diagonal term $a_k b_k$ appears exactly $n-1$ times on the left hand side of (6) (once together with every other index), so

$$\sum_{1 \leq i < j \leq n} (a_i b_i + a_j b_j) = (n-1) \sum_{k=1}^n a_k b_k. \quad (7)$$

Substituting (7) into (6) gives precisely inequality (3):

$$(n-1) \sum_{i=1}^n a_i b_i \geq \sum_{1 \leq i < j \leq n} (a_i b_j + a_j b_i).$$

Thus (3) holds, and by the equivalence established in Step2 we obtain the original Chebyshev inequality (1).

Equality condition. Equality in (4) occurs exactly when $(a_j - a_i)(b_j - b_i) = 0$. Hence for every pair $i < j$ we must have either $a_i = a_j$ or $b_i = b_j$. Because the sequences are non-decreasing, the only way all such products vanish is that ****one entire sequence is constant****:

- either $a_1 = a_2 = \dots = a_n$, or
- $b_1 = b_2 = \dots = b_n$.

The case $n = 1$ is automatically covered (the condition is vacuous). No other configuration can make all pairwise products zero while preserving the ordering, so these are the complete equality cases.

Conclusion. We have rewritten Chebyshev's sum inequality as a sum over unordered pairs, used the elementary non-negativity $(a_j - a_i)(b_j - b_i) \geq 0$ that follows from the similarly ordered sequences, and summed over all index pairs. The argument works for every $n \geq 1$. Equality holds ****iff**** at least one of the sequences is constant (or trivially when $n = 1$). $\square$